Settlement Economy

A quantitative design study: how goods made, used and traded by settlements would behave in Rise of Mankind, measured on 30 real games, and what each implementation choice in the codebase changes

Findings.
1. The settlements of 30 harness games were rebuilt from their telemetry; the rebuilt city counts match the games' own on 45300 of 45334 player-turns.
2. With the game's own yields and governor, the plots those cities work feed their observed populations on all but 6.9 % of city-turns.
3. Treating staple food as a good that cities keep to themselves raises the share of citizens short of food from 9.2 % (today's rule) to 11.5 %; sharing it within a trade network lowers it to 5.1 %; stores lower it to 1.2 %, and spoilage brings it back to 4.2 %.
4. Whole-unit arithmetic throws away 8.8 % of all production to truncation; hundredths lose nothing.
5. Single-precision floating point gave a different allocation for the same goods, depending only on the order in which cities were summed, in 1803 of 120323 rationings: unusable in multiplayer.
6. Needs not matched to what the land can give stay unmet whatever the distribution: fuel and tools are short for most citizens in every variant. A governor that knows the city's needs lowers the fuel shortage from 66.4 % to 53.4 %.
7. Between AIs today, 580 of 1738 deals involve resources, 568 of them swaps of one resource for another and 1 for gold per turn. With goods, players lack every turn what others have spare: on average 740 units of variety food per game alone.
8. The goods pass costs 0.95 ms per turn at 300 cities and 100 goods; the AI's marginal values for every good 98 ms.
9. Buildings, civics and specialists add as much as the land: production is 4.08 times what the worked plots give at turn 150, commerce 4.3 times; and on this map every trade route pays the 1.00 floor before modifiers.

How it was made. Telemetry of the 30 harness games of build 504830a0dee0 (GitLab `main` `3bec049`) that served as the baseline of predictions P18 and P19; the map they were played on; the mod's XML records and DLL source at the same commit. Scripts in `research/economy/settlement/tools`: `games.py` (settlements and deals), `model.py` (the reference model), `calibrate.py`, `analysis.py`, `bench.c` and `sfigures.py`. Every number in the text is read from what they wrote. Parameters that are design choices are marked as assumptions and collected in the appendix. Companion to *Land, not tokens* (research/economy).

1. Questions and sources

Each question is answered from a named source; the model was built only for what no source answers (section 6).

Question	Source
How many settlements does a game have, how large, and how fast do they grow?	30 harness games: per-player cities and population each turn; founding, capture and raze events with their plots
Can the rebuilt settlements be trusted?	Rebuilt city counts against the telemetry's; the food balance of the rebuilt cities against the populations the games fed
How does trade in resources work today, and how do the AIs use it?	CvDeal.cpp (TRADE_RESOURCES), CvStructs.h (TradeData), CvEnums.h (TradeableItems); the deals in the 30 games
How do buildings, civics and trade routes change food, growth, production and commerce, and by how much?	CvCity::getYieldRate, getBaseYieldRateModifier, getBaseTradeProfit; CIV4BuildingInfos.xml, CIV4CivicInfos.xml (bc.py); observed production and commerce rates against the rebuilt plots (yields.py)
What do food and growth do today?	CvCity::foodConsumption, foodDifference, doGrowth; CvPlayer::getGrowthThreshold; GlobalDefines.xml
What would each implementation choice change for these settlements?	The reference model (section 6) run on the rebuilt settlements
What does it cost per turn and in the save?	bench.c, the model's per-turn pass in C, timed; byte counts of its state
Is it deterministic, and does it conserve goods?	The model's state hashed per run; integer accounting; the same rationing in floating point

2. Setup

Scenario. PrivateMaps/Mediterranean, Standard (64 by 44 plots, 1842 land), 10 civilizations, all AI; Normal speed, Noble; no barbarians, no events, no tech diffusion (the scenario file's options); 150 turns; build 504830a0dee0. 30 games, one per seed from `ledger.py seeds P18`, all of which reached turn 150.

The engine's food rules (DLL and XML at 3bec049):

- A citizen eats `FOOD_CONSUMPTION_PER_POPULATION` = 3 food per turn; unhealthiness adds to it (`CvCity::foodConsumption` = population × 3 – health rate).
- A city grows when its stored food reaches `BASE_CITY_GROWTH_THRESHOLD` + size × `CITY_GROWTH_MULTIPLIER` = 30 + 10 × size, scaled by game speed, start era and the AI's handicap, all 100 % here (`CvPlayer::getGrowthThreshold`).
- A city works up to `NUM_CITY_PLOTS` = 21 plots, the fat cross of radius 2. Its own plot yields at least 3 food, 2 production and 1 commerce (`iMinCity`, CIV4YieldInfos.xml).
- Food is local: each city's surplus or deficit changes only its own store (`CvCity::doGrowth`).

3. The games

Over 150 turns a game grows to 100.6 cities on average (10th to 90th percentile 93 to 107) and 671.3 citizens (figure 1).

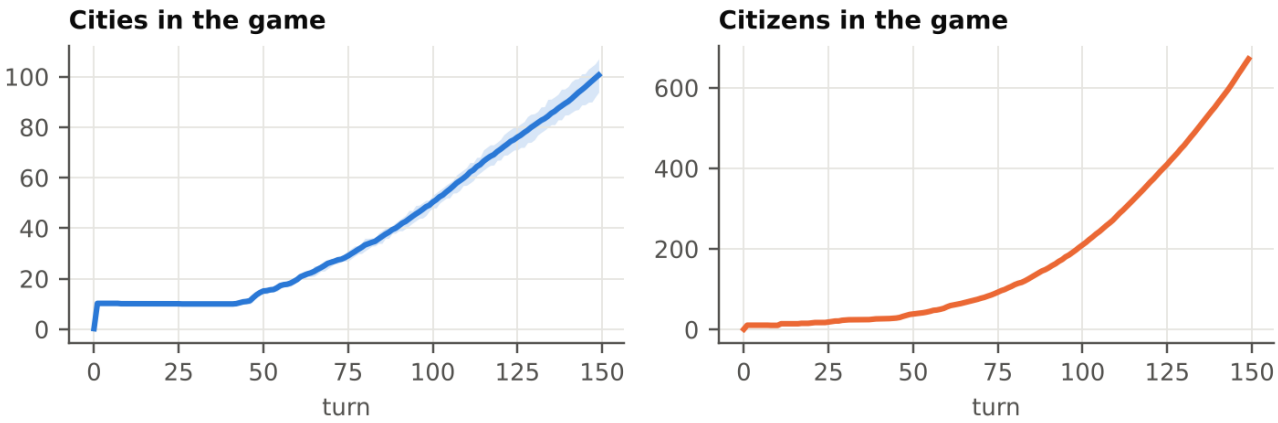


Figure 1. Cities (left; line: mean of the 30 games, band: 10th to 90th percentile) and citizens (right; mean) over the games.

The telemetry gives each player's total population, not each city's. It is split among the player's cities by one of two rules: equally, or in proportion to the square root of each city's age (older cities larger). Every result below is computed under the age rule and checked under the equal rule (section 7.6). Figure 2 shows the city sizes the two rules give.

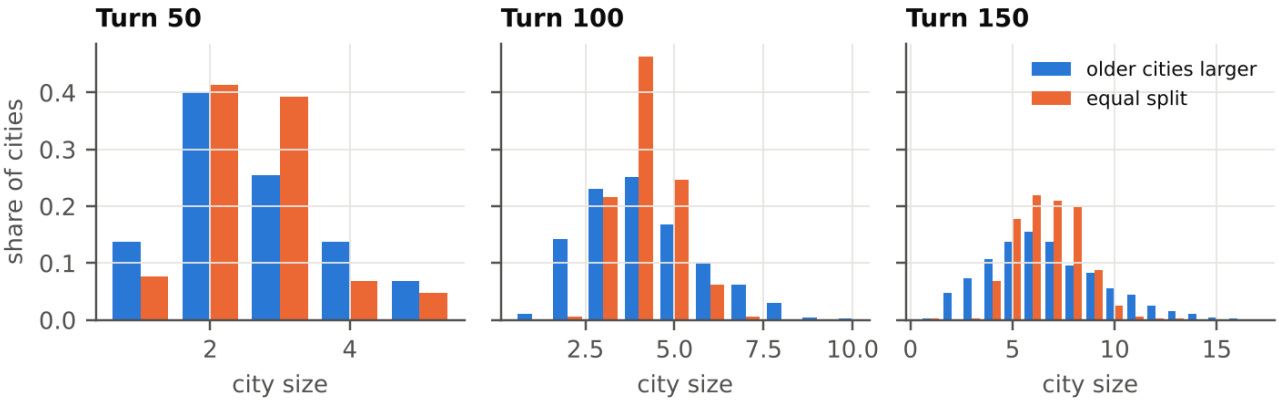


Figure 2. Share of cities by size at turns 50, 100 and 150 under the two splitting rules (first 10 games).

4. Rebuilding the settlements

4.1 Method

Cities are replayed from the events: a founding creates a city on its plot, a capture changes its owner, a razing removes it. Each plot within a city's fat cross belongs to the nearest city, ties to the older one. A city of size *n* works its own plot and *n* of its plots chosen as the game's governor would: by rank (food 3, production 2, commerce 1; assumption), and if those cannot feed the city, the *n* plots richest in food, as `CvCityAI::AI_juggleCitizens` avoids starvation. Plot yields are the XML's: terrain, feature, river, hills (–1 food, +1 production, `iHillsChange`), and the improvements a worker would have built: an irrigated Farm (+2 food) on flat grassland and plains with fresh water, irrigation chains from turn 60 (assumption), a Mine (+1 production) on hills, a Cottage elsewhere, and on the map's resource plots the improvement with the best bonus yield.

4.2 Validation

City counts. The rebuilt cities match the number of cities the telemetry reports for each player on **45300 of 45334** player-turns.

Food. The games fed their populations: a city whose food fell short would have shrunk. So the share of rebuilt city-turns whose worked plots cannot feed the observed population measures how far the rebuilt yields fall short of the real ones. Figure 3 shows it in three stages:

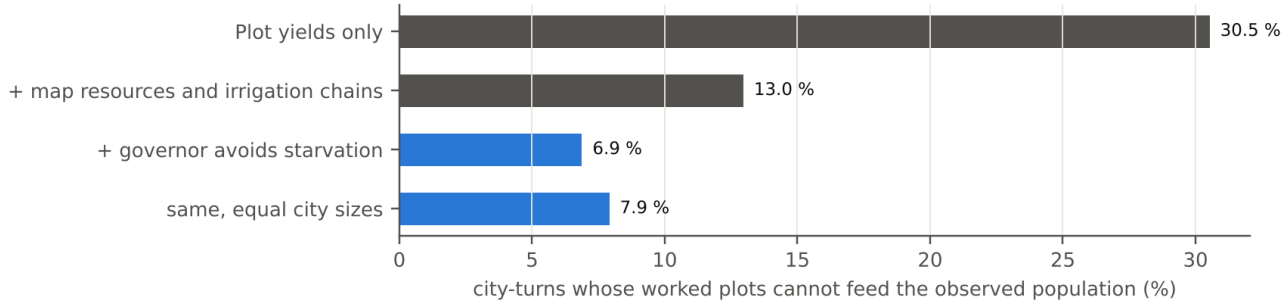


Figure 3. Share of city-turns whose worked plots give less than 3 food per citizen, as the reconstruction adds the map's resources and irrigation chains, then the governor's starvation avoidance. All 30 games.

Plot yields alone leave 30.5 % of city-turns short; adding the map's resources and irrigation chains, 13.0 %; adding the governor's rule, **6.9 %** (7.9 % with equal city sizes). The remaining shortfall is food the reconstruction does not model, such as Lighthouse and Harbor food on water plots and building effects, and brief starvation the real game allowed. It is the baseline against which every variant is compared.

5. Trade, buildings and civics today

5.1 Trade in resources

In the code. A deal item is a `TradeData`: an item type and one integer (`CvStructs.h`), exported to the executable. `TRADE_RESOURCES` moves one copy of a resource for as long as the deal lasts (`CvDeal::startTrade` calls `changeBonusExport` and `changeBonusImport` by 1); either side may cancel after `PEACE_TREATY_LENGTH` = 10 turns (`CvDeal::turnsToCancel`). There is no quantity: a deal gives a copy or nothing. The list of item types, `TradeableItems` (`CvEnums.h`), is compiled into the executable too, with a different numbering on its side (`#ifdef _USRDLL`), so new item types are not a DLL-only change.

In the games. The AIs made 1738 deals. 1136 were open borders for open borders. **580 involved resources**: 568 swaps of one resource for another, 11 resources given for nothing, and 1 for gold per turn. Resource deals lasted a median of 20 turns (321 with a recorded end). Food resources lead: fish 120, wheat 114, apple 110, sugar 84, shrimp 72 (figure 4).

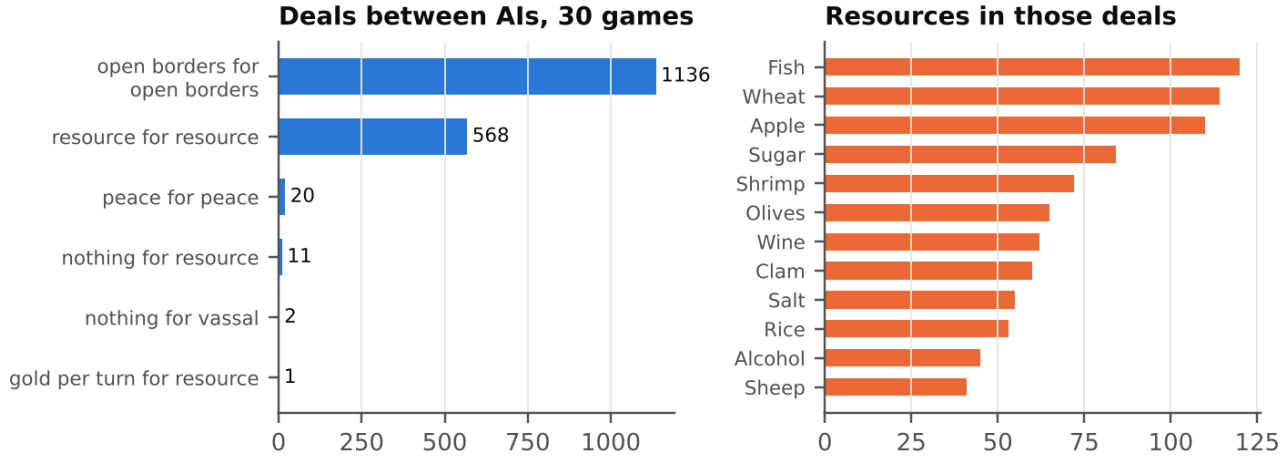


Figure 4. What the AIs traded in the 30 games: deals by content (left), and the resources in them (right).

Between AIs, then, the resource economy is barter of on/off copies, almost never paid for. Paying gold per turn for a copy is how the AI treats the human, which is the case the user describes; it does not arise between AIs at all.

5.2 How buildings and civics change a city's yields

The engine computes every yield of a city the same way (`CvCity::getYieldRate`): a base, the sum of its worked plots, its specialists, the buildings' flat yields and, for commerce, its trade routes, times a modifier that adds up the city's buildings, the resources it holds, power, its landmass, the player's civics and, in the capital, the capital bonus (`CvCity::getBaseYieldRateModifier`):

$$Y = \left\lfloor \frac{B \cdot \max(0,\ 100 + m_{buildings} + m_{resources} + m_{power} + m_{area} + m_{civics} + m_{capital})}{100} \right\rfloor \quad (1)$$

The same formula applies to food, so a civic or building that changes food by a percentage changes growth directly. Growth also depends on how much of the food box a city keeps when it grows (`iFoodKept`, used by `CvCity::doGrowth`).

Effect in CIV4BuildingInfos.xml	Buildings
Food by percent / flat food / food on sea plots	11 / 7 / 4
Food kept after growth	10
Production by percent / with power	15 / 9
Gold or research by percent	71
Trade routes / trade route yield by percent	26 / 30
Health / happiness	28 / 48
Specialist slots	146

Examples from the records (of 389 buildings): Town Essentials (cost 70) adds 3 % food, 1 food on every sea plot, keeps 30 % of the food box after growth, adds 15 % gold, 1 trade route, 3 health and 3 happiness. The Forge adds 25 % commerce and costs 6 health; the Factory adds 15 % production; the Market 20 % gold and 25 % trade route yield; the Harbor 2 trade routes and 50 % trade route yield.

Of 68 civics, 18 change yields by percent, 19 change what trade routes yield, 9 change the yields of improvements and 8 add trade routes. Every civilization starts with Oral Law (food -25 %, production -25 %, commerce -10 %) and Hunter-Gatherer (commerce from trade -50 %). Barter turns trade into food (+20 %); Planned (an Economy civic) into food and production (+50 %, +50 %). The Employment civics add yield to whole classes of improvements: Caste System +2 food to farms, Labor Allocation +2 food to farms, pastures and fishing boats. National Socialism forbids foreign trade.

5.3 Their measured effect in the games

The telemetry records each player's production as the sum of its cities' `getYieldRate(YIELD_PRODUCTION)`, and its gold, research, culture and espionage rates: what the game produced after every building, civic, specialist and trade route. Dividing them by what the rebuilt cities' worked plots give measures, turn by turn, how much everything other than the land adds (figure 5).

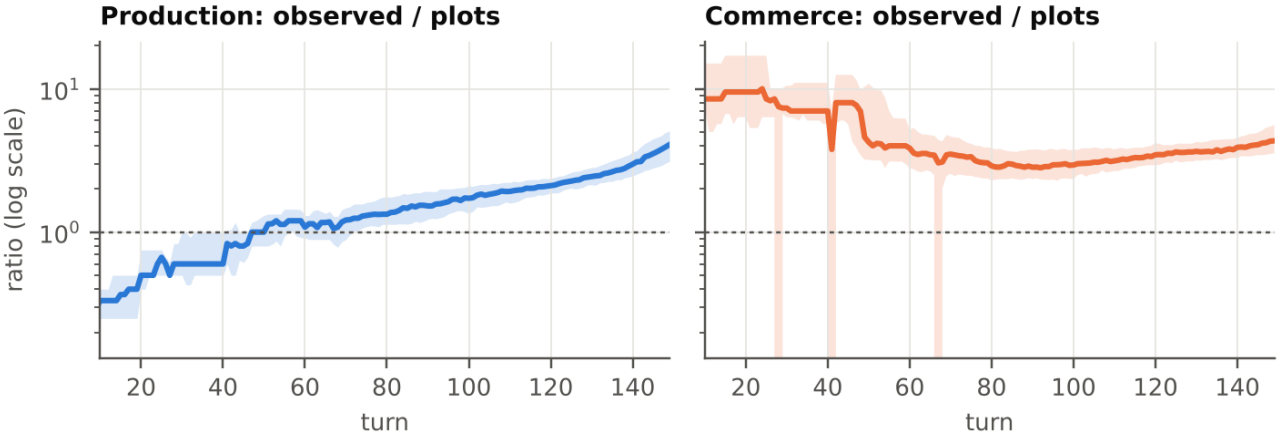


Figure 5. Observed production and commerce divided by what the worked plots give, per player and turn (line: median over the players of the 30 games; band: middle half). Dashed: the plots alone. Logarithmic scale.

Production equals the plots' at turn 50 (median ratio 1.00): Oral Law's -25 % is roughly offset by early buildings. It is 1.73 times the plots at turn 100 and **4.08** at turn 150 (middle half 3.1 to 5.1): by the end, about three quarters of production comes from buildings, civics and specialists, not from land. Commerce is several times the plots' throughout (4.6, 2.9 and 4.3 at turns 50, 100 and 150): the palace, trade routes, specialists and percentage buildings, on top of the little commerce the plots give.

Food under the starting civic. The telemetry has no food rate. If Oral Law's -25 % food held, the rebuilt cities' worked plots could not feed the observed population on **26.9 %** of 173,278 city-turns, against 5.9 % without it. Reading: the AIs leave Oral Law early, or their cities draw food from sources the reconstruction lacks; it would be contradicted if the games' cities kept Oral Law long, which a civics field in the telemetry would show.

5.4 Trade routes

A route's profit is (`CvCity::getBaseTradeProfit`) the smaller of the partner's size times `THEIR_POPULATION_TRADE_PERCENT` (50) and the distance times the world's `iTradeProfitPercent` (50 on a Standard map), times `TRADE_PROFIT_PERCENT` (20 %), at least 1.00, then times the route modifiers (buildings, capital +25 %, overseas +50 %, foreign +100 %) and turned into yields by the civics' trade yield modifiers:

$$\pi = \max\left(1.00,\ \tfrac{20}{100} \cdot \frac{\min(50\,P_{partner},\ 50\,d)}{100}\right) \cdot \frac{100 + m_{route}}{100} \tag{2}$$

For the rebuilt cities, the best domestic partner pays the floor of 1.00 on **435 of 435** routes at turn 50, **1473 of 1473** at turn 100 and **2708 of 3025** at turn 150 (mean 1.02). On a map this size a route is worth 1.00 times its modifiers; its value comes from the number of routes and the modifiers, never from what or whom it connects.

5.5 What this means for a goods economy

Today	In the goods economy	Measured or computed here
Percentage yield modifiers of buildings and civics	The same formula (equation 1) applied to each good's output, with modifiers per good or per category (food, metals, textiles)	Late production is 4.1 times the plots': buildings must remain the main producers
Food kept after growth (Granary, Town Essentials 30 %)	The granary reserve of staple (section 6.5)	Stores cut the staple shortage to 1.2 %
Food on sea plots (Town Essentials +1)	Fish output of worked sea plots	Fish is the good stores lose most (spoilage)
Specialists (146 buildings give slots)	Recipes run by people: a smith makes tools, a weaver cloth	Fuel and tools short in every variant from plots alone
Civics on yields and improvements	Policies on goods: Oral Law -25 % on staple and goods; Caste System +2 staple per farm; Barter turns trade into food	With Oral Law's food the rebuilt cities would starve on 26.9 % of city-turns
Trade routes: 1.00 commerce each, times modifiers	Channels for goods: each route carries a capacity of goods per turn between cities and civilizations	Moving all potential trade needs 0.42 units per route per turn with one route per city (18.3 units per turn per game, 42.9 cities)
No foreign trade (National Socialism)	An embargo on goods with other civilizations	Removes the inter-player flows of section 7.7

6. The reference model

No source says what a goods economy would do to these settlements, so the model was built for it. It applies the proposed rules to the rebuilt cities, turn by turn, without changing the games: populations stay as observed. Every quantity is an integer in hundredths of a unit, as the DLL would hold it; no floating point enters a rule.

6.1 Goods and needs

Need	Goods that meet it (in order)	Per citizen per turn, ancient / classical / medieval
staple	grain, fish	3.00 / 3.00 / 3.00
variety	dairy, fruit	0.20 / 0.35 / 0.50
clothing	cloth, wool	0.10 / 0.15 / 0.20
fuel	coal, wood	0.15 / 0.20 / 0.30
tools	tools	0.00 / 0.03 / 0.06
luxury	wine	0.03 / 0.06 / 0.10

The staple need is 3 per citizen so that one unit of staple is one unit of the game's food. The other levels, the eras (turns 1 to 50, 51 to 100, 101 to 150) and the twelve goods are assumptions chosen for illustration; section 7.3 shows how to set them from the land.

6.2 Production

A worked plot *j* gives goods from its yields and its qualities; for a land plot with food *f*, the grades *G* (0 to 100) of the provinces seeded from the map's resource plots, as in *Land, not tokens*:

$$q_{grain} = 100\,f \text{ (farmed)}, \quad 70\,f \text{ (else)}; \quad q_{dairy} = 30\,f + \frac{G_{sheep}}{2}; \quad q_{wool} = 0.4\,G_{sheep}; \quad q_{fruit} = 0.6\,G_{wine}; \tag{3}$$

$$q_{wood} = 100\,[\text{forest}]; \quad q_{iron,\,copper,\,coal} = G_{iron,\,copper,\,coal}\,[\text{hills}]; \quad q_{fish} = 100\,f + G_{fish}\ \text{(water)} \tag{4}$$

Grain, fruit and dairy are multiplied by a harvest factor of 80 to 120 % per region and turn, drawn from the game's seed by a fixed integer generator (xorshift), so a game's weather is the same every time it is replayed.

6.3 Recipes

Each city runs two recipes up to a capacity $c = 100\,(1 + \blacksquare b/4 \blacksquare)$ hundredths per turn, *b* its share of the player's buildings: tools from iron and fuel (wood first, then coal), and cloth from wool, one for one (assumptions).

$$n_{tools} = \min(c,\ q_{iron},\ q_{wood} + q_{coal}), \qquad n_{cloth} = \min(c,\ q_{wool}) \tag{5}$$

6.4 Distribution and rationing

Within a group of cities (one city, or all cities of a player, the trade network), each need is met from its goods in order. When a good is short, it is shared in proportion to what each city still needs, in exact integers by largest remainders, ties broken by city index:

$$x_i = \left\lfloor \frac{A\,d_i}{\sum_k d_k} \right\rfloor + [\,i \in R\,], \qquad |R| = A - \sum_k \left\lfloor \frac{A\,d_k}{\sum_j d_j} \right\rfloor \tag{6}$$

A is the amount available, d_i city *i*'s demand, and *R* the cities with the largest remainders $A\,d_i \bmod \Sigma d$. The shares sum to *A* exactly. The goods taken are then drawn from the cities that hold them, by the same rule. A variant loses 10 % of what moves between cities.

6.5 Stores, spoilage and growth

Surplus staple food fills the food box, as today; with stores, a granary reserve of three turns' need fills first. Other surpluses are lost without stores; with stores they are kept up to 10 units per citizen per good, and perishable goods lose a share each turn (fish 50 %, dairy 20 %, fruit 10 %, grain 2 %; assumptions):

$$X_{t+1} = \min\left(C,\ X_t + q_t - u_t - \left\lfloor \frac{s\,(X_t + q_t - u_t)}{100} \right\rfloor\right) \tag{7}$$

6.6 Governors

Two rules choose the plots a city works. The **yield governor** is the game's, as rebuilt in section 4. The **goods-aware governor** fills each slot with the plot that adds most, counting its contribution to needs still open (staple twice) plus its ranked yields. It is greedy and deterministic.

6.7 Determinism and conservation

The model's state after each turn is hashed. Run twice on the same game and variant, the hashes are **identical**. All arithmetic is integer, so goods are conserved exactly: everything produced is consumed, stored, spoiled, sent to growth, lost in transport or, without stores, lost.

6.8 The variants

Variant	Staple	Distribution	Stores	Other
Today: food local, goods on top	local food, as today	goods kept in the city	no	
Staple as goods, kept in the city	a good	city	no	
Staple as goods, shared in the network	a good	player network	no	
Shared, 10 % lost in transport	a good	network, 10 % loss	no	
Shared, with stores	a good	network	yes	no spoilage
Shared, stores, spoilage	a good	network	yes	spoilage
Shared, whole units only	a good	network	no	integers in whole units
Shared, needs by city size	a good	network	no	need levels by size, not era
Shared, equal city sizes	a good	network	no	equal population split
Shared, goods-aware governor	a good	network	no	goods-aware governor
Shared, stores, spoilage, goods-aware	a good	network	yes	spoilage, goods-aware governor

7. What each choice changes

7.1 Food: a good or not, local or shared

Figure 6 gives the share of citizens in cities short of staple food under each variant, over all city-turns of the 30 games.

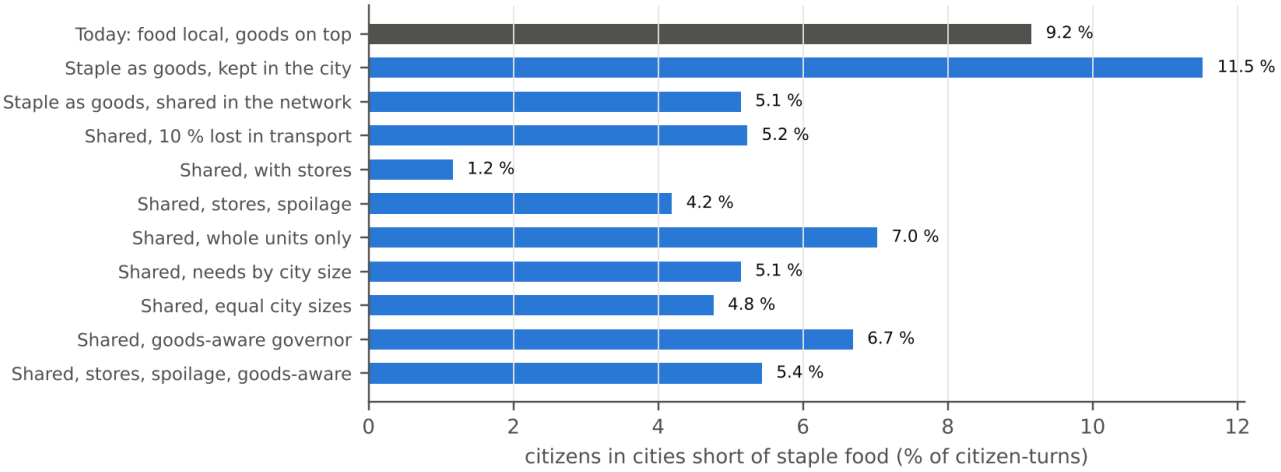


Figure 6. Citizens in cities short of staple food, by variant (% of citizen-turns). Gray: today's rule, the baseline.

Making food a good and keeping it in the city is slightly worse than today (11.5 % against 9.2 %), because the harvest now varies. Sharing it within the network halves the shortage (5.1 %): surpluses of farming cities cover deficits of mining and coastal ones. A 10 % transport loss barely changes that (5.2 %). Stores nearly remove it (1.2 %): the granary reserve absorbs bad harvests. Spoilage takes back part of the gain (4.2 %).

By city size (figure 7), sharing helps the middle sizes most: cities of size 4 to 6 short of staple fall from 8.8 % of city-turns under today's rule to 3.9 % when shared, and 7 to 10 from 13.6 % to 8.1 %.

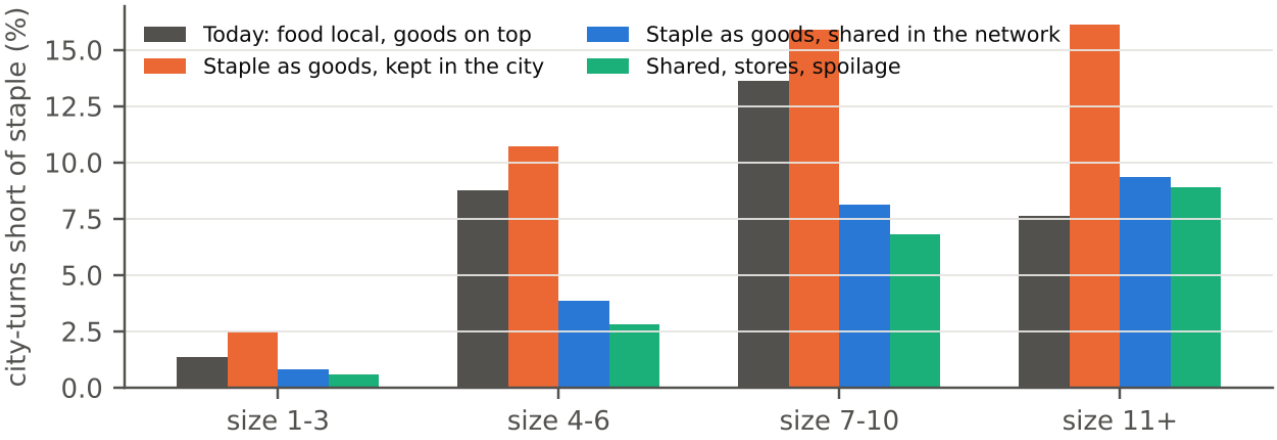


Figure 7. City-turns short of staple food by city size, for four variants.

What this means for the code. Shared staple changes who grows: today a city grows only from its own food (`CvCity::doGrowth`), so food-poor sites stay small. With sharing, a mining town fed by its farming neighbours grows. The network pass must run before the cities' growth, that is in `CvPlayer::doTurn` before the loop that calls `CvCity::doTurn`, and `CvCity::foodDifference` must read the staple the city received instead of its own food yield.

7.2 Needs over the game

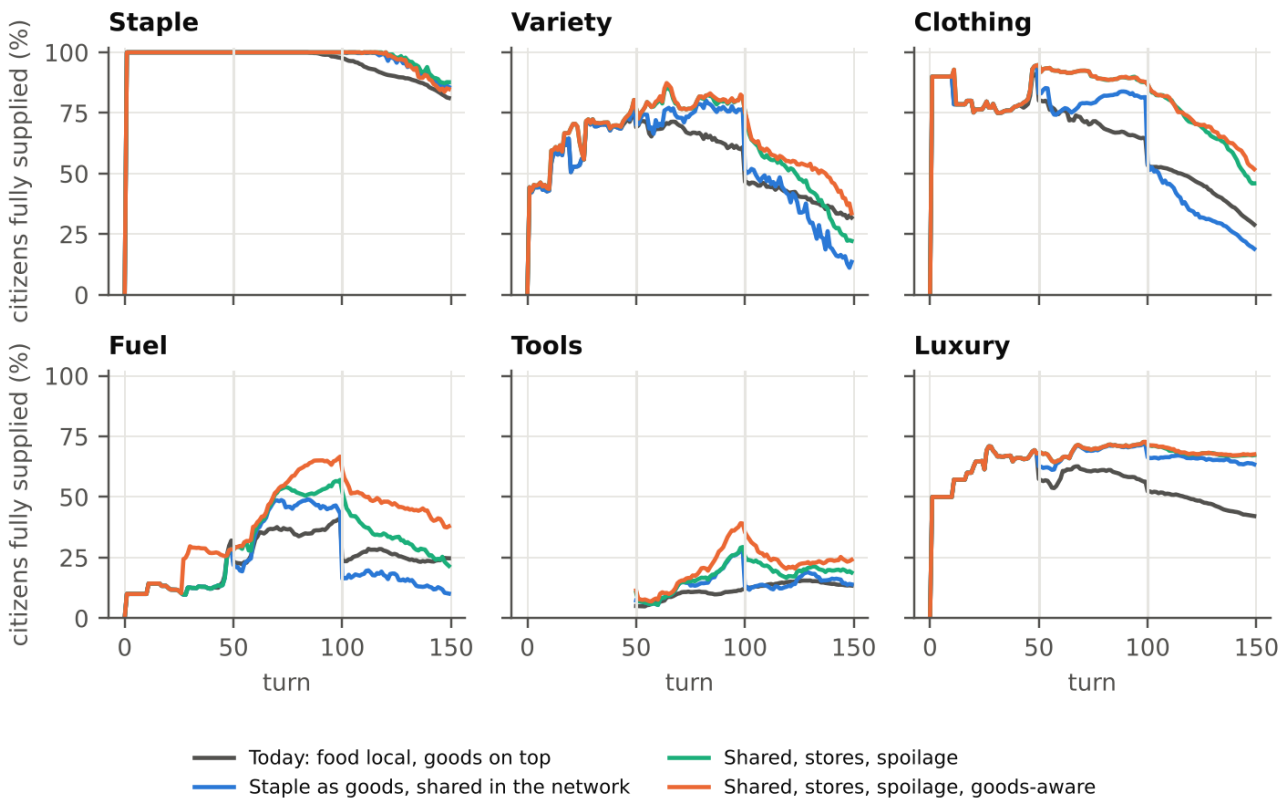


Figure 8. Citizens fully supplied with each need, per turn, for four variants. Tools are needed from the classical era on. Vertical lines: the era boundaries.

With staple shared, 5.1 % of citizen-turns are short of staple; 39.5 % are fully supplied with variety food, 42.0 % with clothing and 66.0 % with luxury; fuel and tools stay short for most citizens in every variant. Stores raise variety, clothing and fuel by carrying surpluses forward. The goods-aware governor lifts fuel (53.4 % short against 66.4 %) and variety (43.5 % against 50.8 %) by working forests and pastures, at a small cost in staple (5.4 % against 4.2 %).

7.3 Setting needs from the land

Whether a need can be met is decided before distribution: by what the land the cities own could give. Figure 9 compares, for each need, what all plots of all cities could give per citizen if every one were worked, with the need per citizen. At turn 150 the land could give 5.49 staple per citizen against a need of 3; for fuel 0.53 against 0.30; for tools 0.10 against 0.06; for luxury 0.17 against 0.10.

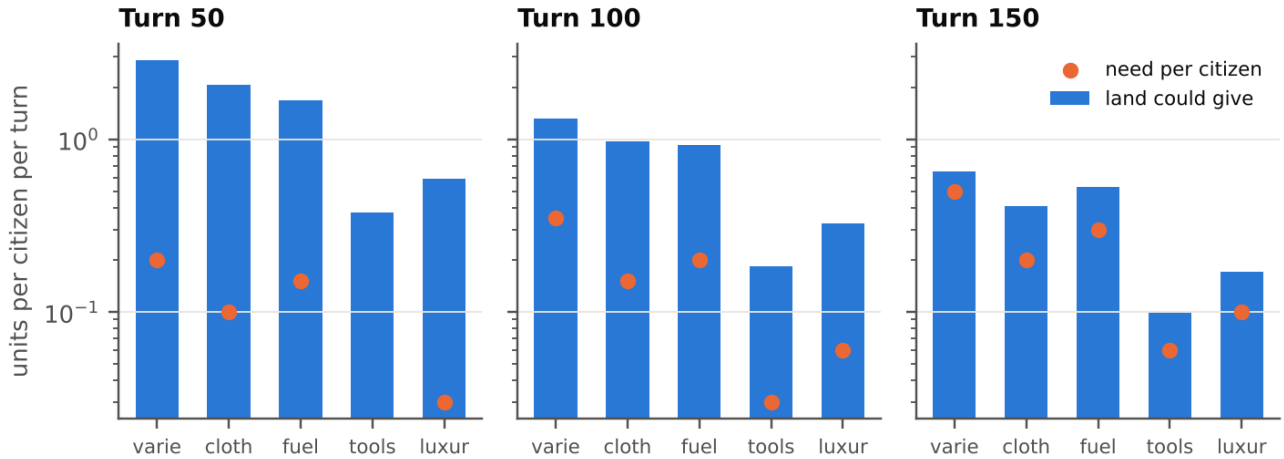


Figure 9. What the land of all cities could give per citizen if every plot were worked (bars), against the need per citizen (dots), at turns 50, 100 and 150. Logarithmic scale.

Design rule. A need level *u* per citizen is sustainable only if the land can give it with the share of plots a city can spare from food. With staple taking 3 per citizen out of a land capacity K_{staple}, the share of worked plots left for a

good g is roughly $1 - 3/K_{staple}$, and

$$u_g \leq \left(1 - \frac{3}{K_{staple}}\right) K_g \tag{8}$$

At turn 150, K_{staple} = 5.49, so a share of 0.45 of the land is spare for other goods, and the bounds are: variety 0.29 against a need of 0.50; clothing 0.19 against a need of 0.20; fuel 0.24 against a need of 0.30; tools 0.04 against a need of 0.06; luxury 0.08 against a need of 0.10. Every need set at its medieval level is above its bound. Such a good must come from trade, from buildings (charcoal, a forge) or from a lower need. As the cities fill their land, K_{staple} falls (18.1 at turn 50, 10.3 at turn 100, 5.5 at turn 150), and with it the room for everything else. The model makes the bound computable on any map before a line of game code is written.

7.4 Stores and spoilage

Waste as a share of everything produced: 7.1 % with goods kept in the city and no stores, 4.5 % shared, 0.8 % with stores, 15.9 % with stores and spoilage, 15.0 % with the goods-aware governor. Stores turn waste into supply; spoilage costs 15.1 points of production but makes perishable goods behave as they should: fish must be eaten where it is landed.

7.5 Units and arithmetic

Whole units. Rounding each plot's goods down to whole units loses **8.8 %** of all production, and the needs met fall with it (staple short 7.0 %, variety short 83.7 %, against 5.1 % and 60.5 % in hundredths). Goods must be held in hundredths, or every small plot yield of a good disappears.

Floating point. The same rationing was repeated in single-precision floating point, summing the cities' demands in two orders. Of **120323** rationings, the total differed between the two orders in **38082** and the allocation to some city in **1803**. In multiplayer each computer must reach the same result bit for bit, or the game goes out of sync. Separately, rounding shares down without the remainder step left goods unallocated in 115854 of the rationings; the largest-remainder step of equation 6 is needed whatever the arithmetic. Integer rationing with largest remainders is exact and order-independent by construction.

7.6 Need tiers and population splits

Need levels by city size instead of era lower the shortages (variety short 35.1 % against 60.5 %, fuel 56.5 % against 79.8 %), because small cities need less. Splitting each player's population equally instead of by age changes the staple shortage from 5.1 % to 4.8 % and variety from 60.5 % to 62.8 %: the conclusions do not depend on the splitting rule.

7.7 What could be traded

Each turn, some players are short of a need that others have spare goods for. Summed over the game, the amount that could change hands, the smaller of the spare supply and the unmet need, is the market goods would create (figure 10).

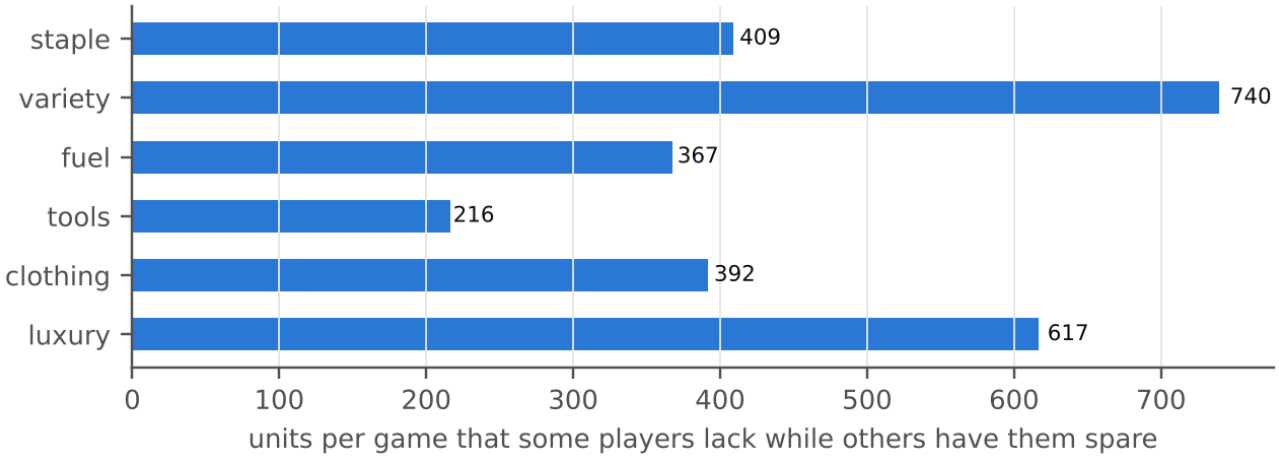


Figure 10. Units per game, by need, that some players lack while others have them spare (shared staple, no stores).

Against today's 580 resource deals in 30 games, mostly swaps of copies, this is a continuous flow every turn of staple 409, variety 740, fuel 367, tools 216, clothing 392, luxury 617. The AI must decide every turn what to sell, at what price and to whom, which is why its valuation (section 8.4) is the centre of the implementation.

7.8 Cost

Figure 11 gives the per-turn cost of the goods pass, written in C as the DLL would run it, for 60 to 600 cities and 12, 40 and 100 goods; and the cost of the AI's marginal values computed by brute force, one recomputation of its network per good.

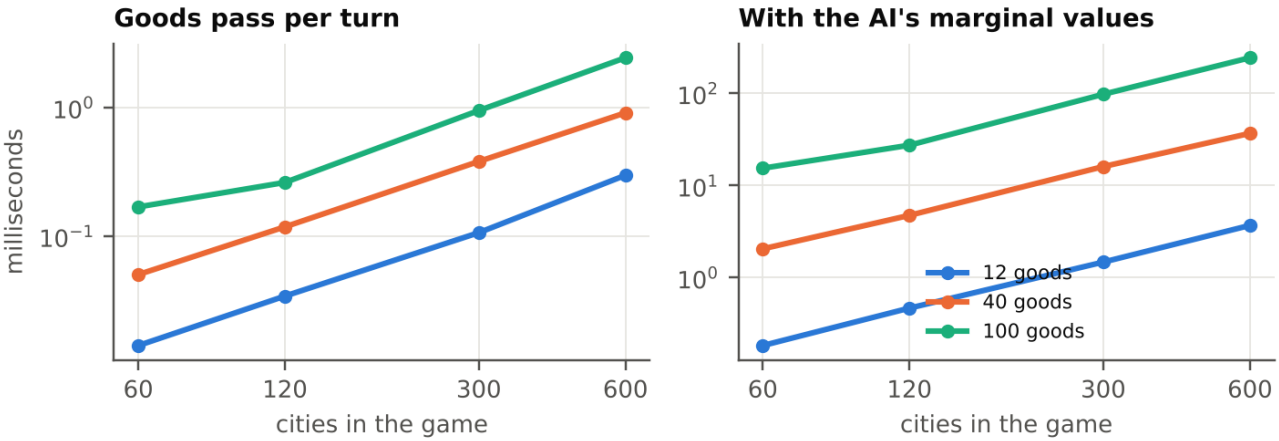


Figure 11. Milliseconds per game turn for the goods pass (left) and with the AI's marginal values (right). Host: Intel Xeon 2.1 GHz, gcc 13 -O2; the game's MSVC 2003 build will differ by a constant factor.

At this scenario's size (120 cities, 12 goods) the pass takes 0.034 ms per turn and 0.5 ms with marginal values. At 300 cities and 100 goods, 0.95 ms and 98 ms. A late turn of a large RoM game takes about 20.8 s (research/graphics); even the largest case is below half a percent of it. The state to save is 300,000 bytes at 300 cities and 100 goods.

8. Implementation in the codebase

8.1 Data

Where	What	Saved
XML: a new CIV4GoodInfos.xml, or new tags on CIV4BonusInfos.xml	Unit, category, spoilage, storage, base value; needs per citizen by era or size	no
XML: CIV4BuildingInfos.xml	Recipes: inputs and outputs per run, capacity; storage capacity	no
XML: CIV4UnitInfos.xml	Goods cost per unit; supply use per turn in the field	no
CvPlot	Province grades and reserves (Land, not tokens)	yes, with a version flag
CvCity	Per good: stock, last turn's production, use and receipts (int, hundredths); per need: share met	yes: CvCity::read and write take a uiFlag
CvPlayer	Contracts in force; prices if markets are used	yes

8.2 The turn

- **CvPlayer::doTurn**, before the loop over `CvCity::doTurn`: for each trade network (the engine's `CvPlotGroup`), collect production, run recipes in dependency order, deliver contracts, ration needs, update stocks.
- **CvCity::foodDifference**: the staple received replaces the city's own food yield, so `doGrowth` keeps working unchanged.
- **CvCity::changeFreeBonus**: a city supplied with a good holds its resource count, so building and unit requirements and happiness rules read goods without change (section 2 of Land, not tokens).
- **CvCity::getProductionNeeded** and the unit builder: goods costs drawn from the city's stock when a unit is completed.
- **Randomness** (harvests): the game's synchronised generator, `getSorenRandNum`, never the system's.

8.3 Trade

Three ways to carry goods in deals, from least to most engine risk:

Option	How	Risk
Reuse TRADE_RESOURCES	One item per unit per turn of a good: a contract of 5 wheat per turn is five items. The screen shows them as it shows resources today.	Low; clumsy for large quantities
Pack the quantity	m_iData = good × 65,536 + quantity, interpreted by the DLL. Whether the executable's screen can show such an item's text is open.	Medium; needs a test in the game
A Python trade screen	Contracts, shipments, seeds and packages in a screen of RoM's own, sending its result to the DLL.	Low for the engine; most interface work

8.4 The AI

The AI needs one quantity everywhere: the marginal value of a unit of each good to each of its networks, the change in its weighted needs met and production when that good's supply rises by one unit. Section 7.8 shows that computing it by brute force every turn is affordable. It then drives: the governor's choice of plots (section 6.6), building choice (recipes whose outputs are worth more than their inputs), what to offer and accept in trade, and when to go to war over a province. This generalises the valuation of merge request !63, which values today's resources by their effect in the AI's own cities.

8.5 Saves and multiplayer

New fields go behind the version flag each class already reads first (`CvCity::read` reads `uiFlag`); old saves load with empty stocks. All rules are integer and use the synchronised generator, so the game stays in sync; section 7.5 shows why floating point cannot be used.

9. Recommendations

Decision	Recommended	Evidence
Scale and arithmetic	Integers in hundredths; one staple unit = one food	Whole units lose 8.8 % of production; floats changed allocations in 1803 rationings
Food	Staple is a good, shared in the network, feeding the existing food box	Staple shortage 5.1 % shared against 11.5 % local
Catalogue	25 to 40 goods	Cost is negligible at any size; the limit is AI and interface, not CPU
Need tiers	By city size, rising with era	Variety short 35.1 % by size against 60.5 % by era
Need levels	Set from the land's capacity (7.3) per map type	Fuel and tools short in every variant at the assumed levels
Stores and spoilage	Both, from the start	Stores: staple short 4.2 %; spoilage keeps fish local
Distribution	Automatic within a network; transport loss optional	Loss of 10 % changes staple shortage by 0.1 points
Governor	Goods-aware	Fuel short 53.4 % against 66.4 %
Buildings and civics	Keep the modifier formula; give it per-good modifiers; specialists run recipes	Late production 4.1 times the plots'
Trade routes	Goods channels with a capacity per route, beside their commerce	All routes at the 1.00 floor; 0.42 units per route per turn would carry all trade
Trade	Contracts as repeated TRADE_RESOURCES items first; a Python screen later	TradeData has no quantity; the item list is the executable's
Order of work	Goods and needs in cities, then trade, then industry and units	Trade and industry need the goods accounts and the AI's marginal values

10. How the implementation would be judged

Under the repository's rules: each step registered as a prediction before it is built; safety claims (no crash, not slower) by 5 seeded harness runs per arm; claims that play gets better by at least 30; the human's experience by the user's games. The model gives each step a number to predict before the game code exists: for example, that with shared staple and stores fewer than 4.2 % of citizen-turns in this scenario are short of food, measured by the harness from the real cities instead of the rebuilt ones.

11. Limits

- Populations are taken from the games, not simulated: the model says what goods would do to these cities, not how the cities would have grown differently. Feeding the model's growth back needs the rules in the DLL and the harness.
- City sizes are split from player totals by an assumed rule; section 7.6 shows the results do not hinge on it.
- The reconstruction leaves 6.9 % of city-turns short of food under today's rules (section 4.2); every comparison is against that baseline.
- Need levels, recipes, spoilage and storage are illustrative; the method, not these values, is the result.
- One scenario, all AI, 150 turns: late-game industry is not reached.
- The goods model counts plots only; section 5.3 shows buildings, civics and specialists add as much again by turn 150, and they enter the goods economy as modifiers and recipes not simulated here.

Appendix: parameters, data and fingerprints

Parameter	Value	Kind
Staple per citizen	3 (FOOD_CONSUMPTION_PER_POPULATION)	fact
Growth threshold	30 + 10 × size	fact
Plot ranking	food 3, production 2, commerce 1	assumption
Irrigation chains	from turn 60	assumption
Harvest factor	80 to 120 % per region (16 by 11 plots) and turn	assumption
Province widths	ores 1.5 plots, wine and livestock 2, fish 1.5	assumption
Spoilage per turn	grain 2 %, fish 50 %, dairy 20 %, fruit 10 %	assumption
Storage	10 units per citizen per good; granary 3 turns of staple	assumption
Transport loss	10 % (one variant)	assumption
Recipe capacity	1 + buildings per city / 4 units per turn	assumption

Fingerprints (first 16 hex digits of the hash of every state of every game, per variant): today_food `2c9a37a3a58fb44c`; goods_local `3fa9722432e66a03`; goods_network `ca8ad5a374fc7e06`; goods_network_loss `1d37cb09ded06366`; network_store `36377075f1ae3bc3`; network_store_spoil `e884c9c6ec943676`; whole_units `69eb937294fa95ed`; tiers_size `411fd693bf20266d`; equal_split `d98a0663b71f5ee4`; aware_network `0f00aa3038060ed6`; aware_network_store_spoil `7031d347945dd3fd`.

Data. research/economy/settlement/data: games.json (settlements and deals of the 30 games), facts_medstd.json (records and map), model.json, series.json, calibration.json, deals.json, bench.csv, snumbers.json.